

4

DERIVATIVE OF PARAMETRIC FUNCTIONS

If $x = f(t)$ and $y = g(t)$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{g'(t)}{f'(t)}$$

$$\text{And } \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx} = \frac{d}{dt} \left(\frac{dy}{dx} \right) / \frac{dx}{dt}$$

5

DIFFERENTIATION OF A FUNCTION WITH RESPECT TO ANOTHER FUNCTION

If $y = f(x)$ and $z = g(x)$, then in order to find the derivative of $f(x)$ w.r.t. $g(x)$, we use the formula

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$$

6

LOGARITHMIC DIFFERENTIATION

Properties of Logarithms

- $\log_e(mn) = \log_e m + \log_e n$
- $\log_e \left(\frac{m}{n} \right) = \log_e m - \log_e n$
- $\log_e (m)^n = n \log_e |m|$
- $\log_e e = 1$
- $\log_n m = \frac{\log_e m}{\log_e n}$
- $\log_n m \cdot \log_m n = 1$

Shorter Methods of Finding the Derivative of a Logarithmic Function

If $y = [f(x)]^{g(x)}$, then to find $\frac{dy}{dx}$, in addition to the method discussed above, we can also apply any of the following two methods:

Method 1

Step 1. Express $y = [f(x)]^{g(x)} = e^{g(x) \log f(x)}$
 $[\because a^x = e^{x \log a}]$

Step 2. Differentiate w.r.t. x to obtain $\frac{dy}{dx}$

Method 2

Step 1. Evaluate

A = Differential coefficient of y treating $f(x)$ as constant.

Step 2. Evaluate

B = Differential coefficient of y treating $g(x)$ as constant.

Step 3. $\frac{dy}{dx} = A + B$.

7

DIFFERENTIATION OF INVERSE TRIGONOMETRIC FUNCTIONS

Important Substitutions to Reduce the Function to a Simpler Form

Expressions

$$\sqrt{a^2 - x^2}$$

$$\sqrt{x^2 - a^2}$$

$$\sqrt{a^2 + x^2}$$

$$\frac{a-x}{a+x} \text{ or } \frac{a+x}{a-x}$$

$$\sqrt{\frac{a-x}{a+x}} \text{ or } \sqrt{\frac{a+x}{a-x}}$$

$$\sqrt{\frac{a^2 - x^2}{a^2 + x^2}} \text{ or } \sqrt{\frac{a^2 + x^2}{a^2 - x^2}}$$

Substitutions

Put $x = a \sin \theta$ or $x = a \cos \theta$

Put $x = a \sec \theta$ or $x = a \operatorname{cosec} \theta$

Put $x = a \tan \theta$ or $x = a \cot \theta$

Put $x = a \tan \theta$

Put $x = a \cos \theta$

Put $x^2 = a^2 \cos \theta$

8

DIFFERENTIATION OF A FUNCTION GIVEN IN THE FORM OF A DETERMINANT

$$\text{If } y = \begin{vmatrix} f(x) & g(x) & h(x) \\ p(x) & q(x) & r(x) \\ u(x) & v(x) & w(x) \end{vmatrix}, \text{ then}$$

$$\frac{dy}{dx} = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ p(x) & q(x) & r(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ p'(x) & q'(x) & r'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ p(x) & q(x) & r(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

Note: The differentiation of a determinant can be done in columns also.

APPLICATION OF DERIVATIVES

(1) Common application of derivatives.

- Finding Rate of Change of a Quantity
- Finding the Approximation Value
- Finding the equation of a Tangent and Normal To a Curve
- Finding Maxima and Minima, and Point of Inflection
- Determining Increasing and Decreasing Functions

(2) Approximations

Assume we have a function $y = f(x)$, which is defined in the interval $[a, a + h]$, then the average rate of change in the function in the given interval is $(f(a + h) - f(a))/h$.
Now using the definition of derivative, we can write

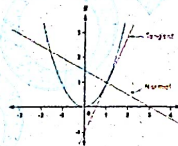
$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

which is also the instantaneous rate of change of the function $f(x)$ at a . Now, for a very small value of h , we can write
 $f'(a) \approx (f(a+h) - f(a))/h$

(3) Equations of tangent & Normal

- Equation of tangent at (x_1, y_1) : $y - y_1 = \left(\frac{dy}{dx}\right)_{(x_1, y_1)} (x - x_1)$
- Equation of Normal at (x_1, y_1) :

$$y - y_1 = \left(\frac{-dx}{dy}\right)_{(x_1, y_1)} (x - x_1)$$



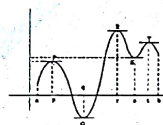
(4) Increasing decreasing functions.

Properties of monotonic functions :

- (1) if $f(x)$ is continuous on $[a, b]$ such that $f'(c) \geq 0$ for each c , then $f(x)$ is monotonically increasing function. Similar definition goes for monotonically decreasing function.
- (2) if $f(x)$ is strictly increasing function on $[a, b]$ then $f^{-1}(x)$ exists & is also strictly increasing on $[a, b]$. Similar result follows for strictly decreasing functions.
- (3) if $f(x)$ & $g(x)$ are two continuous & differentiable functions, then we can relate $\log f(x)$ & $\log g(x)$ by the following table

	$f(x)$	$g(x)$	$\log f(x)$
+ denotes increasing function	+	+	+
- denotes decreasing function	-	-	+
	+	-	-
	-	+	-

(5) Maxima & Minima



Point P, R, T are points of local maxima
Point Q, S are point of local minima

Global maxima, global minima

$$\text{Maximum} = \max \{f(a), f(c_1), f(c_2), \dots, f(c_n), f(b)\}$$

$$\text{Minimum} = \min \{f(a), f(c_1), f(c_2), \dots, f(c_n), f(b)\}$$

$c_1, c_2, \dots, c_n$ are n critical points.

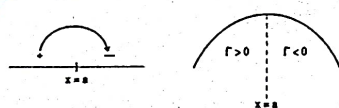
(6) Test Of Local Maxima & Minima

First Derivative Test:

- Step 1: Find the critical points of the function by putting $f'(x) = 0$
Step 2: For each of the critical points obtained in step 1 do the following :

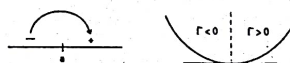
Case 1: $x = a$ is local maxima

if $f'(x)$ changes from + to - as x passes through a

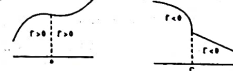


Case 2: $x = a$ is local minima

if the sign of $f'(x)$ changes from - to + as x passes through a

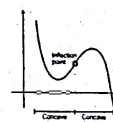


Case 3: There is no sign change across a this means that $x = a$ is neither a point of maxima nor minima.



(7) Point of inflection

A point of inflection is point where the curve changes its shape from convex to concave or from concave to convex.



(8) Higher Order Test

Let f be a differentiable function on interval I . Let c be any point in the domain of f such that

$$(1) f'(c) = f''(c) = f'''(c) = \dots = f^{(n-1)}(c) = 0 \text{ and}$$

$$(2) f^{(n)}(c) \neq 0 \text{ and } n \text{ is even}$$

then f has a local maximum at c if $f^{(n)}(c) < 0$ and a local minimum at c if $f^{(n)}(c) > 0$.

$f^{(n)}(c) < 0 \Rightarrow x = c$ is a local maxima
 $f^{(n)}(c) > 0 \Rightarrow x = c$ is a local minima

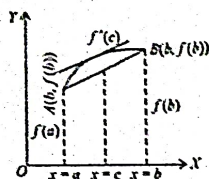
(10) Lagrange's Mean Value Theorem

If a function f defined on the closed interval $[a, b]$, is

1. continuous on $[a, b]$ and

2. derivable on (a, b) , then there exists atleast one real number c between a and b ($a < c < b$) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Geometrical interpretation

The theorem states that between two points A and B on the graph of f there exists atleast one point where the tangent is parallel to the chord AB.

(9) Rolle's Theorem

If a function f defined on the closed interval $[a, b]$, is

1. continuous on $[a, b]$,

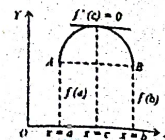
2. derivable on (a, b) and

3. $f(a) = f(b)$, then there exists atleast one real number c between a and b ($a < c < b$) such that $f'(c) = 0$.

Geometrical interpretation

Let the curve $y = f(x)$, which is continuous on $[a, b]$ and derivable on (a, b) , be drawn.

The theorem states that between two points with equal ordinates on the graph of f , there exists atleast one point where the tangent is parallel to x -axis.



Algebraic interpretation

Between two zeros a and b of $f(x)$ (i.e., between two roots a and b of $f(x) = 0$) there exists atleast one zero of $f'(x)$.

Key Points to Remember

1. The value of c may not be unique i.e., there can be more than one such c .
2. Every polynomial function is continuous and differentiable for all real x .
3. The function $\log x$ is continuous on $(0, \infty)$.
4. $|x - a|$ is not differentiable at $x = a$ (e.g., $|x|$ is not differentiable at $x = 0$).
5. If the derivative of a function has finite and unique value on an interval, then the function is derivable on that interval.

(11) Parametric coordinates

$$(1) x^{2/3} + y^{2/3} = a^{2/3} : x = a \cos^3 \theta, y = a \sin^3 \theta$$

$$(2) \sqrt{x} + \sqrt{y} = \sqrt{a} : x = a \cos^4 \theta, y = a \sin^4 \theta$$

$$(3) \frac{x^a}{a^a} + \frac{y^a}{b^a} = 1 : x = a (\cos \theta)^{2/a}, y = b (\sin \theta)^{2/a}$$

$$(4) c^2(x^2 + y^2) = x^2 y^2 : x = c \sec \theta, y = c \csc \theta$$

$$(5) y^2 = x^3 : x = t^2, y = t^3$$

(12) Angle of intersection of two Curves

The angle between the tangents to the two curves at their point of intersection.

Let c_1 & c_2 be two curves.

$$m_1 = \tan \theta_1 = \left(\frac{dy}{dx} \right)_{c_1}$$

$$\text{Angle of intersection, } \theta = \tan^{-1} \left| \frac{\left(\frac{dy}{dx} \right)_{c_1} - \left(\frac{dy}{dx} \right)_{c_2}}{1 + \left(\frac{dy}{dx} \right)_{c_1} \left(\frac{dy}{dx} \right)_{c_2}} \right|$$

$$m_2 = \tan \theta_2 = \left(\frac{dy}{dx} \right)_{c_2}$$

(13) Orthogonal Curves

If the angle of intersection of two curves is a right angle, the two curves are said to be orthogonal. If the curves are orthogonal,

$$\left(\frac{dy}{dx} \right)_{c_1} \left(\frac{dy}{dx} \right)_{c_2} = -1$$

(14) Subtangent & Subnormal

$$\text{Length of Tangent} = \left| y_1 \sqrt{1 + \left(\frac{dx}{dy} \right)_{x_1, y_1}^2} \right|$$

$$\text{Length of Normal} = \left| y_1 \sqrt{1 + \left(\frac{dy}{dx} \right)_{x_1, y_1}^2} \right|$$

$$\text{Length of Subtangent} = \left| y_1 \left(\frac{dx}{dy} \right)_{x_1, y_1} \right|$$

(Projection of tangent)

$$\text{Length of subnormal} = \left| y_1 \left(\frac{dy}{dx} \right)_{x_1, y_1} \right|$$

(Projection of normal)

(15) Leibnitz-rule

$$\frac{d}{dx} \left[\int_{\phi(x)}^{\psi(x)} f(t) dt \right] = f(\psi(x)) \left\{ \frac{d}{dx} \psi(x) \right\} - f(\phi(x)) \left\{ \frac{d}{dx} \phi(x) \right\}$$

(16) Extrema of discontinuous Functions

1. Minimum of discontinuous Functions :

For Minimum at $x = a$

$$f(a) \leq f(a + h)$$

$$f(a) \leq f(a - h)$$

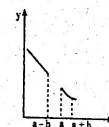
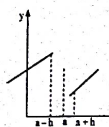
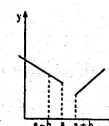
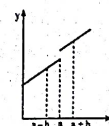
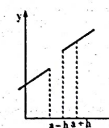
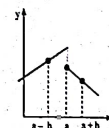
2. Maximum of discontinuous Functions:

For maximum at $x = a$

$$f(a) \geq f(a + h)$$

$$f(a) \geq f(a - h)$$

3. Neither Maximum nor minimum exists:



Indefinite Integral

Let $f(x)$ be a function, the family of all its primitives (or antiderivatives) is called the Indefinite Integral of $f(x)$ and is denoted by $\int f(x)dx$.

Standard Integrals

$$\begin{aligned}
 (i) \int x^n dx &= \frac{x^{n+1}}{n+1} + C, n \neq -1 & (ii) \int \frac{1}{x} dx &= \log|x| + C & (iii) \int e^x dx &= e^x + C & (iv) \int a^x dx &= \frac{a^x}{\log a} + C & (v) \int \sin x dx &= -\cos x + C & (vi) \int \cos x dx &= \sin x + C \\
 (vii) \int \sec^2 x dx &= \tan x + C & (viii) \int \csc^2 x dx &= -\cot x + C & (ix) \int \sec x \tan x dx &= \sec x + C & (x) \int \csc x \cot x dx &= -\csc x + C & (xi) \int \cot x dx &= \log|\sin x| + C \\
 (xii) \int \tan x dx &= \log|\sec x| + C & (xiii) \int \csc x dx &= \log|\csc x - \cot x| + C & (xiv) \int \sec x dx &= \log|\sec x + \tan x| + C & (xv) \int \frac{1}{\sqrt{a^2 - x^2}} dx &= \sin^{-1}\left(\frac{x}{a}\right) + C \\
 (xvi) \int \frac{1}{\sqrt{a^2 + x^2}} dx &= \cos^{-1}\left(\frac{x}{a}\right) + C & (xvii) \int \frac{1}{a^2 + x^2} dx &= \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C & (xviii) \int \frac{1}{x\sqrt{x^2 - a^2}} dx &= \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C & (xix) \int \frac{1}{a^2 + x^2} dx &= \frac{1}{a} \cot^{-1}\left(\frac{x}{a}\right) + C \\
 (xx) \int \frac{1}{x\sqrt{x^2 - a^2}} dx &= \frac{1}{a} \csc^{-1}\left(\frac{x}{a}\right) + C
 \end{aligned}$$

Integration By Substitution

Expression	Substitution
$a^2 + x^2$	$x = a \tan \theta$ or $a \cot \theta$
$a^2 - x^2$	$x = a \sin \theta$ or $a \cos \theta$
$x^2 - a^2$	$x = a \sec \theta$ or $a \csc \theta$
$\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$	$x = a \cos 2\theta$
$\sqrt{\frac{x-a}{\beta-x}}$ or $\sqrt{(x-a)(x-\beta)}$	$x = a \cos^2 \theta + \beta \sin^2 \theta$

Integration Using Partial Fractions

$$\begin{aligned}
 (i) \frac{px+q}{(x-a)(x-b)} &= \frac{A}{x-a} + \frac{B}{x-b}, a \neq b & (ii) \frac{px+q}{(x-a)^2} &= \frac{A}{x-a} + \frac{B}{(x-a)^2} \\
 (iii) \frac{px^2+qx+r}{(x-a)(x-b)(x-c)} &= \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c} \\
 (iv) \frac{px^2+qx+r}{(x-a)^2(x-b)} &= \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b} \\
 (v) \frac{px^2+qx+r}{(x-a)(x^2+bx+c)} &= \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}
 \end{aligned}$$

where $x^2 + bx + c$ cannot be factorised further.

Integration By Parts

$$\int u \cdot v dx = u \int v dx - \int \left[\frac{du}{dx} \cdot \int v dx \right] dx$$

Follow ILATE

I = Inverse Trigonometric
 L = Logarithmic
 A = Algebraic
 T = Trigonometric
 E = Exponential

5.

QUIK LOOK

$$\begin{aligned}
 \int \frac{dx}{x^2 - a^2} &= \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C & \int \frac{dx}{x^2 + a^2} &= \log|x + \sqrt{x^2 + a^2}| + C & \int \sqrt{x^2 - a^2} dx &= \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log|x + \sqrt{x^2 - a^2}| + C \\
 \int \frac{dx}{a^2 - x^2} &= \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C & \int e^x [f(x) + f'(x)] dx &= e^x f(x) + C & \int \sqrt{a^2 - x^2} dx &= \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C \\
 \int \frac{dx}{\sqrt{x^2 - a^2}} &= \log|x + \sqrt{x^2 - a^2}| + C & \int [xf'(x) + f(x)] dx &= xf(x) + C & \int \sqrt{x^2 + a^2} dx &= \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log|x + \sqrt{x^2 + a^2}| + C
 \end{aligned}$$

Integrals of different forms:

$$(1) \int \sin^m x dx, \int \cos^n x dx \text{ where } m \leq 4$$

express $\sin^m x$ and $\cos^n x$ in terms of sines and cosines of multiples of x by using the following identities:

$$(i) \sin^2 x = \frac{1 - \cos 2x}{2} \quad (ii) \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$(iii) \sin 3x = 3 \sin x - 4 \sin^3 x \quad (iv) \cos 3x = 4 \cos^3 x - 3 \cos x$$

$$(2) \int \sin mx \cos nx dx, \int \sin mx \sin nx dx, \int \cos mx \cos nx dx$$

use the following trigonometrical identities:

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B); 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B); 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$(3) \int \frac{f'(x)}{f(x)} dx = \log|f(x)| + C \quad (4) \int \{f(x)\}^n f'(x) dx = \frac{\{f(x)\}^{n+1}}{n+1}, n \neq -1$$

$$(5) \int \tan^m x \sec^{2n} x dx, \int \cot^m x \csc^{2n} x dx; m, n \in \mathbb{N}$$

$$\text{Put } \tan x = t \text{ and } \sec^2 x dx = dt$$

$$(6) \int \sin^m x \cos^n x dx, m, n \in \mathbb{N}$$

If the exponent of $\sin x$ is an odd positive integer put $\cos x = t$

If the exponent of $\cos x$ is an odd positive integer put $\sin x = t$

6.

(9) $\int \sin^m x \cos^n x dx$, where $m, n \in \mathbb{Z}$, $m+n$ is a negative even integer.

Express the integrand in terms of $\tan x$ and $\sec x$ by dividing numerator and denominator by $\cos^2 x$, where $k = -(m+n)$ then put $\tan x = t$.

$$(9) \int \frac{px+q}{ax^2+bx+c} dx$$

To evaluate this,

$$px+q = \lambda \left\{ \frac{d}{dx} (ax^2+bx+c) \right\} + \mu \text{ i.e. } px+q = \lambda(2ax+b) + \mu$$

$$(11) \int \frac{P(x)}{ax^2+bx+c} dx \text{ where } P(x) \text{ is a polynomial of degree two or more}$$

to evaluate this, write $\int \frac{P(x)}{ax^2+bx+c} dx = \int Q(x) dx + \int \frac{R(x)}{ax^2+bx+c} dx$

$$(13) \int \frac{1}{a \sin x + b \cos x} dx, \int \frac{1}{a + b \sin x} dx, \int \frac{1}{a + b \cos x} dx$$

To evaluate this, put $\sin x = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)}$ and $\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$ and simplify.

$$(15) \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2} (a \sin bx - b \cos bx) + C$$

$$(8) \int \frac{1}{ax^2+bx+c} dx$$

express ax^2+bx+c as the sum or difference of two squares.

$$(10) \int (px+q)\sqrt{ax^2+bx+c} dx$$

In order to evaluate this, write

$$px+q = \lambda \frac{d}{dx} (ax^2+bx+c) + \mu \text{ i.e. } px+q = \lambda(2ax+b) + \mu$$

$$(12) \int \frac{1}{a \sin^2 x + b \cos^2 x} dx, \int \frac{1}{a + b \sin^2 x} dx$$

$$\int \frac{1}{a + b \cos^2 x} dx, \int \frac{1}{(a \sin x + b \cos x)^2} dx, \int \frac{1}{a + b \sin^2 x + c \cos^2 x} dx$$

To evaluate this type of integrals, divide numerator and denominator both by $\cos^2 x$.

$$(14) \int \frac{a \sin x + b \cos x}{c \sin x + d \cos x} dx$$

To evaluate this, write Numerator = $\lambda(\text{Diff. of denominator}) + \mu(\text{Denominator})$

$$(16) \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2+b^2} (a \cos bx + b \sin bx) + C$$

$$(17) \int \frac{1}{(ax+b)\sqrt{cx+d}} dx$$

$$cx+d=t^2$$

$$(18) \int \frac{1}{(ax^2+bx+c)\sqrt{px+q}} dx$$

$$\text{put } px+q=t^2$$

$$(19) \int \frac{1}{(ax+b)\sqrt{px^2+qx+r}} dx$$

$$\text{put } ax+b = \frac{1}{t}$$

$$(20) \int \frac{1}{(ax^2+b)\sqrt{cx^2+d}} dx$$

$$\text{put } x = \frac{1}{t} \text{ to obtain}$$

$$(21) \int \frac{-tdt}{(a+bt^2)\sqrt{c+dt^2}}$$

$$\text{substitute } c+dt^2=u^2$$

Reduction formulas

(1) Reduction Formula for Exponential Functions

$$\int x^n e^{mx} dx = \left[\frac{1}{m} x^n e^{mx} \right] - \left[\frac{n}{m} \int x^{n-1} e^{mx} dx \right]$$

$$\int e^{mx} / x^n dx = - \left[e^{mx} / (n-1) x^{n-1} \right] + \left[\frac{m}{n-1} \int e^{mx} / x^{n-1} dx \right], n \neq 1$$

(3) Reduction Formula for Logarithmic Functions

$$\int x^n \ln^m x dx = \frac{x^{n+1} \ln^m x}{n+1} - \frac{m}{n+1} \int x^n \ln^{m-1} x dx$$

$$\int \frac{\ln^m x}{x^n} = - \frac{\ln^m x}{(n-1)x^{n-1}} + \frac{m}{n-1} \int \frac{\ln^{m-1} x}{x^n} dx, n \neq 1$$

(4) Reduction Formula for Inverse Trigonometric Functions

$$\int x^n \arcsin x dx = \frac{x^{n+1} \arcsin x}{n+1} - \frac{1}{n+1} \int \frac{x^{n+1}}{\sqrt{1-x^2}} dx$$

$$\int x^n \arccos x dx = \frac{x^{n+1} \arccos x}{n+1} + \frac{1}{n+1} \int \frac{x^{n+1}}{\sqrt{1-x^2}} dx$$

$$\int x^n \arctan x dx = \frac{x^{n+1} \arctan x}{n+1} - \frac{1}{n+1} \int \frac{x^{n+1}}{1+x^2} dx$$

(2) Reduction Formula for Trigonometric Functions

$$\int \sin^n(x) dx = \frac{-\sin^{n-1}(x) \cos(x)}{n} + \frac{n-1}{n} \int \sin^{n-2}(x) dx$$

$$\int x^n \cos(x) dx = x^n \sin(x) - n \int x^{n-1} \sin(x) dx$$

$$\int x^n \sin(x) dx = -x^n \cos(x) + n \int x^{n-1} \cos(x) dx$$

$$\int \sin^n(x) \cos^m(x) dx = \frac{\sin^{n+1}(x) \cos^{m-1}(x)}{n+m} + \frac{m-1}{n+m} \int \sin^n(x) \cos^{m-2}(x) dx$$

$$\int \frac{dx}{\sin^n x} = - \frac{\cos x}{(n-1) \sin^{n-1} x} + \frac{(n-2)}{(n-1)} \int \frac{dx}{\sin^{n-2} x}, n \neq 1$$

$$\int \frac{dx}{\cos^n x} = \frac{\sin x}{(n-1) \cos^{n-1} x} + \frac{(n-2)}{(n-1)} \int \frac{dx}{\cos^{n-2} x}, n \neq 1$$

$$\int \tan^n(x) dx = \frac{\tan^{n-1}(x)}{n-1} - \int \tan^{n-2}(x) dx$$

(5) Reduction Formula for Algebraic Functions

$$\int \frac{dx}{(ax^2 + bx + c)^n} = \frac{-2ax - b}{(n-1)(b^2 - 4ac)(ax^2 + bx + c)^{n-1}} + \frac{2(2n-3)a}{(n-1)(b^2 - 4ac)} \int \frac{dx}{(ax^2 + bx + c)^{n-1}}, n \neq 1$$

$$\int \frac{dx}{(x^2 + a^2)^n} = \frac{x}{2(n-1)a^2(x^2 + a^2)^{n-1}} + \frac{2n-3}{2(n-1)a^2} \int \frac{dx}{(x^2 + a^2)^{n-1}}, n \neq 1$$

$$\int \frac{dx}{(x^2 - a^2)^n} = \frac{x}{2(n-1)a^2(x^2 - a^2)^{n-1}} - \frac{2n-3}{2(n-1)a^2} \int \frac{dx}{(x^2 - a^2)^{n-1}}, n \neq 1$$

8.

Derived substitutions:

A. Algebraic Twins

$$\int \frac{2x^2}{x^4 + 1} dx = \int \frac{x^2 + 1}{x^4 + 1} dx + \int \frac{x^2 - 1}{x^4 + 1} dx,$$

$$\int \frac{2}{x^4 + 1} dx = \int \frac{x^2 + 1}{x^4 + 1} dx - \int \frac{x^2 - 1}{x^4 + 1} dx$$

$$\int \frac{2x^2}{(x^4 + 1 + kx^2)} dx = \int \frac{2}{(x^4 + 1 + kx^2)} dx$$

B. Trigonometric twins

$$\int \sqrt{\tan x} dx, \int \sqrt{\cot x} dx,$$

$$\int \frac{1}{(\sin^4 x + \cos^4 x)} dx, \int \frac{1}{(\sin^6 x + \cos^6 x)} dx, \int \frac{\pm \sin x \pm \cos x}{a + b \sin x \cos x} dx.$$

Definite Integral

1. Definite Integral As The Limit Of A Sum

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} [f(a) + f(a+h) + \dots + f(a+(n-1)h)]$$

$$h = \frac{b-a}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

The above expression is known as the definite integral as the limit of a sum.

2. Properties of Definite Integrals

$$(i) \int_a^b f(x) dx = \int_a^b f(t) dt$$

$$(vi) \int_a^a f(x) dx = 2 \int_a^a f(x) dx$$

If $f(x)$ is an even function

$$(ii) \int_a^b f(x) dx = - \int_b^a f(x) dx \text{ In particular } \int_a^a f(x) dx = 0$$

$$(vii) \int_{-a}^a f(x) dx = 0,$$

If $f(x)$ is an odd function

$$(iii) \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

$$(viii) \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$$

$$(iv) \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$(ix) \int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(2a-x) = f(x)$$

$= 0$, if $f(2a-x) = -f(x)$

$$(v) \int_a^b f(x) dx = \int_a^b f(a-x) dx$$

3. Walli's formula

$$\int_0^{\pi/2} \sin^n x dx = \int_0^{\pi/2} \cos^n x dx = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdot \dots \cdot \frac{2}{3}, & \text{when } n \text{ is odd} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdot \dots \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}, & \text{when } n \text{ is even} \end{cases}$$

$$\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{(m-1)(m-3) \dots (n-1)(n-3) \dots}{(m+n)(m+n-2) \dots (2 \text{ or } 1)}$$

[If m, n are both odd positive integers or one odd positive integer]

$$\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{(m-1)(m-3) \dots (n-1)(n-3) \dots \pi}{(m+n)(m+n-2) \dots (2 \text{ or } 1) \cdot 2}$$

[If m, n are both even positive integers]

4. Periodic Properties

If $f(x)$ is a periodic function with period T , then

$$01 \int_0^{nT} f(x) dx = n \int_0^T f(x) dx, n \in \mathbb{Z}$$

$$02 \int_a^{a+nT} f(x) dx = n \int_a^T f(x) dx, n \in \mathbb{Z}, a \in \mathbb{R}$$

$$03 \int_m^{m+nT} f(x) dx = (n-m) \int_0^T f(x) dx, m, n \in \mathbb{Z}$$

$$04 \int_{nT}^{(n+1)T} f(x) dx = \int_0^T f(x) dx, n \in \mathbb{Z}, a \in \mathbb{R}$$

5. Advance properties

$$\psi(x) \leq f(x) \leq \phi(x) \text{ for } a \leq x \leq b \text{ then } \int_a^b \psi(x) dx \leq \int_a^b f(x) dx \leq \int_a^b \phi(x) dx$$

$$\text{If } m \leq f(x) \leq M \text{ for } a \leq x \leq b, \text{ then } m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

If $f(x) \geq 0$ on $[a, b]$ then

$$\int_a^b f(x) dx \geq 0$$

6. Leibnitz Theorem

$$F(x) = \int_{g(x)}^{h(x)} f(t) dt, \text{ then}$$

$$\frac{dF(x)}{dx} = h'(x)f(h(x)) - g'(x)f(g(x))$$

Gamma function

$$\Gamma n = \int_0^\infty e^{-x} x^{n-1} dx$$

where n is a positive rational number

Properties of gamma function

$$1) \Gamma(0) = \infty, \Gamma(1) = 1$$

$$2) \Gamma(n+1) = n\Gamma(n)$$

$$3) \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$4) \Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}, 0 < n < 1$$

Beta function

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

NOTE The relationship between beta & gamma function will be

$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

8.

Important results

$$\sum_{r=1}^n r = \frac{n(n+1)}{2}$$

$$\sum_{r=1}^n r^2 = \frac{1}{6} n(n+1)(2n+1)$$

$$\sum_{r=1}^n r^3 = \frac{n^2(n+1)^2}{4}$$

$$\text{In GP, sum of } n \text{ terms, } S_n = \begin{cases} \frac{a(r^n - 1)}{r - 1}, & |r| > 1 \\ an, & r = 1 \\ \frac{a(1 - r^n)}{1 - r}, & |r| < 1 \end{cases}$$

$$\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + (n-1)\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)} \sin(\alpha + (n-1)\beta/2).$$

$$\cos \alpha + \cos(\alpha + \beta) + \dots + \cos(\alpha + (n-1)\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)} \cos(\alpha + (n-1)\beta/2)$$

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \frac{1}{6^2} + \dots \infty = \frac{\pi^2}{12}$$

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \infty = \frac{\pi^2}{6}$$

$$\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \dots \infty = \frac{\pi^2}{24}$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$1/1^2 + 1/3^2 + 1/5^2 + 1/7^2 + \dots \infty = \pi^2/8$$

9

Average Value Theorem

If f is a continuous function on $[a, b]$, then its average value on $[a, b]$ is given by the formula

$$f_{\text{AVG}[a,b]} = \frac{1}{b-a} \int_a^b f(x) dx$$

Application Of Integrals

1

$\int_a^b f(x) dx \neq$ Area under the curve $f(x)$ from a to b

$\int_a^b f(x) dx =$ Algebraic area under the curve $f(x)$ from a to b

2

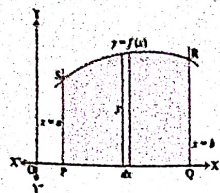
POSITIVE AND NEGATIVE AREA

Area is always taken as positive. If some part of the area lies in the positive side i.e., above x -axis and some part lies in the negative side i.e., below x -axis then the area of two parts should be calculated separately and then add their numerical values to get the desired area.

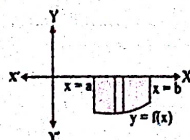
3

Area Under Simple Curves

(i) Area of the region bounded by a curve $y = f(x)$ and



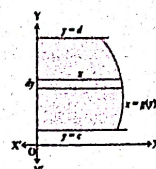
$$\text{Area, } A = \int_a^b dA = \int_a^b y dx = \int_a^b f(x) dx$$



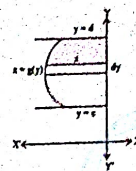
If the position of the curve under consideration is below the x -axis. Then, area is negative. So, we take its absolute value, i.e.,

$$\text{Area}(A) = \left| \int_a^b f(x) dx \right|$$

(ii) Area of the region bounded by a curve $x = f(y)$ and between the two ordinates



$$\text{Area, } A = \int_a^b x dy = \int_a^b f(y) dy$$



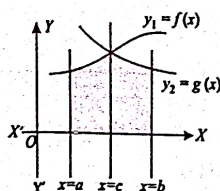
If the position of the curve under consideration is below the y -axis. Then, area is negative. So, we take its absolute value, i.e.,

$$\text{Area}(A) = \left| \int_a^b f(y) dy \right|$$

4

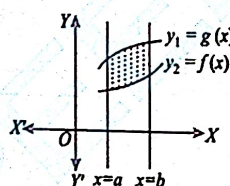
Area Under Different Curves

CASE-I



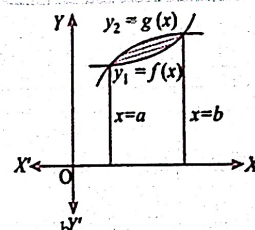
$$A = \int_a^b f(x) dx + \int_c^b g(x) dx$$

CASE-II



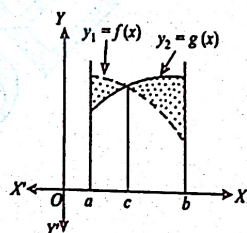
$$A = \int [g(x) - f(x)] dx$$

CASE-III



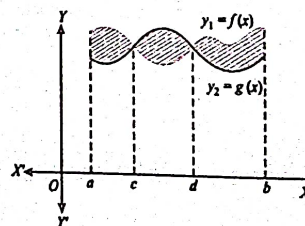
$$A = \int_a^b [g(x) - f(x)] dx$$

CASE-IV



$$A = \int_a^c [f(x) - g(x)] dx + \int_c^b [g(x) - f(x)] dx$$

CASE-V



$$A = \int_a^c (y_1 - y_2) dx + \int_c^d (y_2 - y_1) dx + \int_d^b (y_1 - y_2) dx$$

DIFFERENTIAL EQUATION

1 Order Of Differential Equation

The order of a differential equation is the order of the highest derivative occurring in the differential equation. For example

$\frac{d^2y}{dx^2} + y = 0$ is a second order differential equation

$\left(\frac{d^3y}{dx^3}\right) + x^2\left(\frac{d^2y}{dx^2}\right) = 0$ is a third order differential equation.

2 Degree Of Differential Equation

The degree of a differential equation is the highest degree of the highest derivative occurring in the differential equation when it is a polynomial of the differential coefficients i.e., differential coefficients free from radicals & fractions.

For example

Since, $\frac{d^3y}{dx^3} + x^2\left(\frac{d^2y}{dx^2}\right) = 0$ as order = 3

its degree = 1, as $\frac{d^3y}{dx^3}$ has power 1.

3 Differential Equations With Variable Separable

$$\frac{dy}{dx} = h(y) \cdot g(x)$$

Separating the variables, we have $\frac{dy}{h(y)} = g(x) \cdot dx$

Integrate both sides $\int \frac{dy}{h(y)} = \int g(x) \cdot dx$

4 Reducible to the variable separable type

$\frac{dy}{dx} = f(ax + by + c)$ is solved by putting $ax + by + c = t$, etc

5 Homogeneous differential equation

(i) $P(x, y)dx + Q(x, y)dy = 0$ is called homogenous, if P & Q are Homogeneous functions of the same degree on x & y. Reducible to $y' = f\left(\frac{y}{x}\right)$

substitute $y = xu$, u is unknown function. The equation is transformed to an equation with variable separables.

(ii) $\frac{dy}{dx} = f\left(\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}\right)$, $a_1b_2 - a_2b_1 \neq 0$, then substitute $x = u + h$, $y = v + k$ if $a_1b_2 - a_2b_1 = 0$, $u = ax + b$, y transforms into a variable separable form.

* $P(x, y)$ function is Homogeneous of degree n, if for any real t, $P(tx, ty) = t^n(P(x, y))$

* A differential equation of the form $\frac{dy}{dx} = f(x, y)$ is homogeneous, if $f(x, y)$ is a homogeneous function of degree zero i.e., $f(tx, ty) = t^0 f(x, y)$

6 Exact differential equation

$M(x, y)dx + N(x, y)dy = 0$ is exact if its LHS expression is the exact differential of some function $u(x, y)$.

$du = Mdx + Ndy$

Then solution is $u(x, y) = c$.

The sufficient condition for the differential $Mdx + Ndy = 0$ to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

The solution of $Mdx + Ndy = 0$ is

$\int_{y=\text{constant}} Mdx + \int (\text{terms of } N \text{ not containing } x)dy = c$, provided

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

7 Linear Differential Equations

A differential equation of the form $\frac{dy}{dx} + Py = Q$

where P & Q are constants or functions of only x, is known as a First Order Linear Differential Equation.

$$y(IF) = \int (Q \times IF)dx + c, \quad IF = e^{\int Pdx}$$

$$\frac{dx}{dy} + P'x = Q'$$

$$IF = e^{\int P'dy}$$

$$x(IF) = \int (Q' \times IF)dy + c$$

8 Bernoulli's equation

$\frac{dy}{dx} + Py = Qy^n$ dividing $y^n \rightarrow y^{-n} \frac{dy}{dx} + Py^{1-n} = Q \dots (i)$

$$y^{1-n} = z$$

$$(i) \frac{dz}{dx} + (1-n)Pz = (1-n)Q$$

Solution is

$$ze^{\int (1-n)Pdx} = \int \{(1-n)Q\} \cdot e^{\int (1-n)Pdx} dx$$

9 Solution by inspection

$$(1) d(xy) = xdy + ydx$$

$$(2) d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$$

$$(3) d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}$$

$$(4) d\left(\frac{x^2}{y}\right) = \frac{2xydx - x^2dy}{y^2}$$

$$(5) d\left(\frac{y^2}{x}\right) = \frac{2xydy - y^2dx}{x^2}$$

$$(6) d\left(\frac{x^2}{y^2}\right) = \frac{2xy^2dx - 2x^2ydy}{y^4}$$

$$(7) d\left(\frac{y^2}{x^2}\right) = \frac{2x^2ydy - 2xy^2dx}{x^4}$$

$$(8) d\left(\tan^{-1} \frac{x}{y}\right) = \frac{ydx - 2dy}{x^2 + y^2} = \frac{d(x/y)}{1 + (x/y)^2}$$

$$(9) d\left(\tan^{-1} \frac{y}{x}\right) = \frac{xdy - ydx}{x^2 + y^2} = \frac{d(y/x)}{1 + (y/x)^2}$$

$$(10) d[\ln(xy)] = \frac{xdy + ydx}{xy}$$

$$(11) d\left[\ln\left(\frac{x}{y}\right)\right] = \frac{ydx - xdy}{xy}$$

$$(12) d[\ln(y/x)] = \frac{xdy - ydx}{xy}$$

$$(13) d\left[1/2 \ln(x^2 + y^2)\right] = \frac{xdx + ydy}{x^2 + y^2}$$

$$(14) d(-1/xy) = \frac{xdy + ydx}{x^2y^2}$$

$$(15) d\left(\frac{e^x}{y}\right) = \frac{ye^x dx - e^x dy}{y^2}$$

$$(16) d\left(\frac{e^y}{x}\right) = \frac{xe^y dy - e^y dx}{x^2}$$

$$(17) \frac{d[f(x, y)]^{1-n}}{1-n} = \frac{f'(x, y)}{(f(x, y))^n}$$